Completely impure : 50% y / 50% n => $H(S) = 1$ bit

Completely pure : 100% y / 0% n

or 0% y / 100% n => $H(S) = 0$ bit

Information gain.

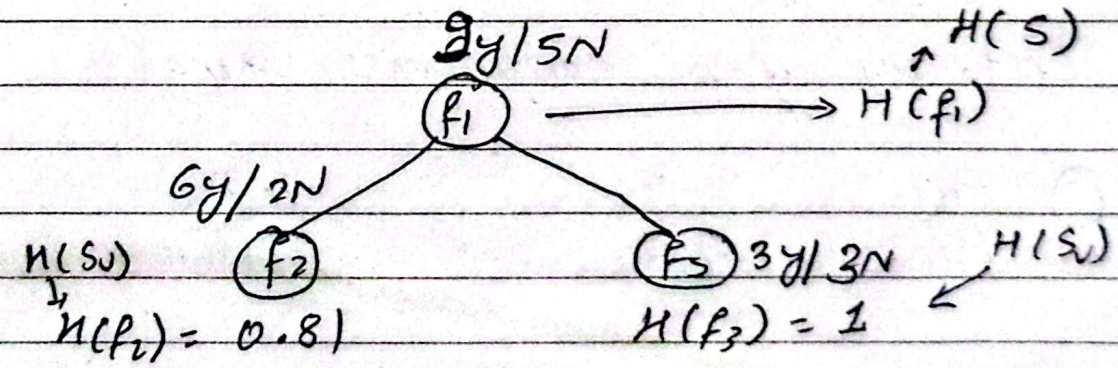
- It measures the expected reduction in entropy.
- It calculate the average of all the entropy

Information gain = E (Parent) − {Weighted Avg} * E(Children)

$$Gain(S, A) = \underbrace{H(S)}_{\text{entropy for especific}} - \sum_{v \in values} \frac{|S_v|}{|S|} \underbrace{H(S_v)}_{\text{entropy after splitting.}}$$

S = subset ~~after~~

S_v = subset : after splitting.



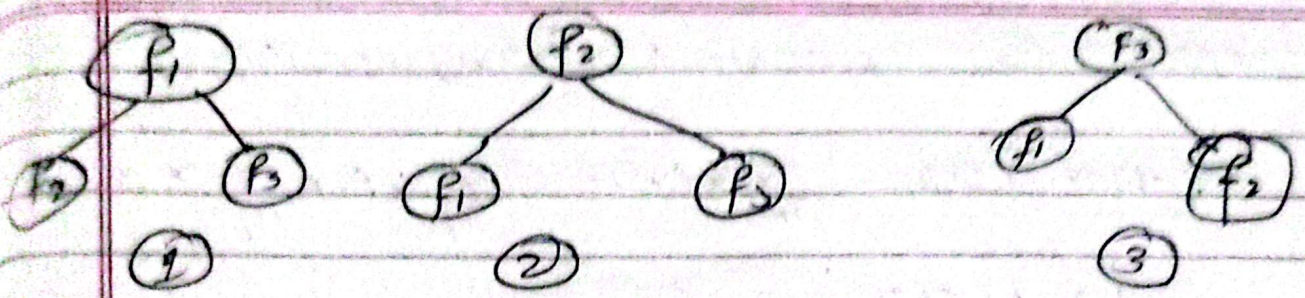
$$H(S) = H(f_1) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$$

$$= 0.94$$

$$Gain(S, f_1) = H(S) - \frac{8}{14} H(f_2) - \frac{6}{14} H(f_3)$$

$$= 0.94 - \frac{8}{14} \times 0.81 - \frac{6}{14} \times 1$$

$$= 0.049$$



- Check the information gain of ① ② & ③, then choose higher one.

Outlook	Temp	Humidity	Windy	Play Tennis.
				No
				Yes
				No

Step 1:

Entropy of parent:

$$E(P) = -P_y \log_2(P_y) - P_y \log_2(P_n)$$

$$= -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right).$$

$$= 0.94$$

Step 2:

outlook

Sunny	Overcast	Rain
3N / 2Y	5Y / 0N	2N / 3Y
$E(S) = 0.97$	$E(O) = 0$	$E(R) = 0.97$

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+ Step3: Calc weighted Entropy of children

Weighted Entropy = $\frac{5}{14} \times 0.97 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.97$

W. E(children) = 0.69

- P(overcast) is a leaf node as it's entropy is 0

+ Step4:

I.G = E(Parent) - {Weight Avg} * E(children)

= 0.97 - 0.69

= 0.28

So, IG (decrease in entropy / Impurity) when you split this data on the basis of Outlook column is : 0.28.

+ Step5:

calculate information gain for all the columns

Higher I.G will be selected to split the ~~column~~ data.

+ Step 6:

Find I.G recursively.